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SUPERFICIAL SAFETY ALIGNMENT HYPOTHESIS

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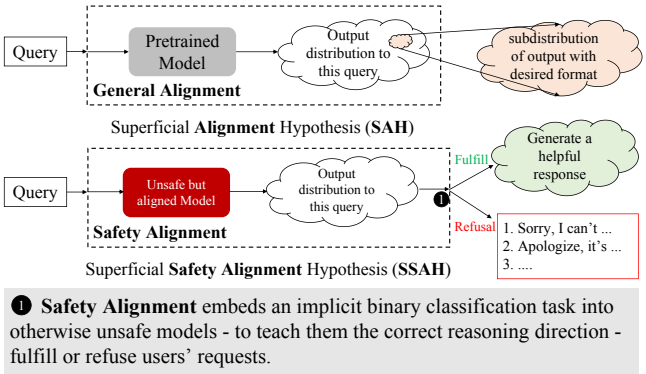
ABSTRACT

As large language models (LLMs) are overwhelmingly more and more integrated into various applications, ensuring they generate safe responses is a pressing need. Previous studies on alignment have largely focused on general instruction-following but have often overlooked the distinct properties of safety alignment, such as the brittleness of safety mechanisms. To bridge the gap, we propose the **Superficial Safety Alignment Hypothesis (SSAH)**, which posits that safety alignment teaches an otherwise unsafe model to choose the correct reasoning direction - fulfill or refuse users' requests - interpreted as an implicit binary classification task. Through **SSAH**, we hypothesize that only a few essential components can establish safety guardrails in LLMs. We successfully identify four types of attribute-critical components: Safety Critical Unit (**SCU**), Utility Critical Unit (**UCU**), Complex Unit (**CU**), and Redundant Unit (**RU**). Our findings show that freezing certain safety-critical components during fine-tuning allows the model to retain its safety attributes while adapting to new tasks. Similarly, we show that leveraging redundant units in the pre-trained model as an "alignment budget" can effectively minimize the alignment tax while achieving the alignment goal. All considered, this paper concludes that the atomic functional unit for safety in LLMs is at the neuron level and underscores that safety alignment should not be complicated. We have code implementation and other information on the project website: https://ssa-h.github.io/.

1 INTRODUCTION

Large language models (LLMs) are demonstrating remarkable capabilities across a broad spectrum of natural language tasks, ranging from text generation to answering complex questions (Achiam et al., 2023; Touvron et al., 2023a;b; Dubey et al., 2024). However, as these models are increasingly integrated into real-world applications, concerns about the risk of generating harmful, unsafe, or unethical content have grown (Askell et al., 2021; Bai et al., 2022; Zeng et al., 2024). This has led to a pressing need for safety alignment, which ensures that LLM outputs are not only coherent and informative but also aligned with human values, ethical standards, and safety considerations.



❶ **Safety Alignment** embeds an implicit binary classification task into otherwise unsafe models - to teach them the correct reasoning direction - fulfill or refuse users' requests.

Figure 1: Superficial **Safety Alignment** Hypotheses

Previous work on alignment has primarily focused on enhancing LLMs' ability to follow general instructions without enough attention to model safety. This trend of treating safety alignment as a subset of general alignment has obscured its distinct challenges (Ouyang et al., 2022; Rafailov et al.,

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2024; Zhou et al., 2024; Liu et al., 2023a; Yuan et al., 2023; Liu et al., 2023b). One major issue is the brittleness of the current safety mechanisms. Despite using benign data during model finetuning, Qi et al. (2023); Yang et al. (2023) have shown that safety mechanisms can fall apart when models are adapted to new tasks. This failure mode amplifies the brittle nature of current safety alignment techniques. At the same time, safety alignment also comes with a tax - a trade-off where improving safety may reduce the model's overall utility or downstream task performances (Ouyang et al., 2022; Wang et al., 2024; Lin et al., 2024). Also, current safety alignment approaches typically require full model finetuning (Hu et al., 2021; Zhang et al., 2023; Thakkar et al., 2024), which is computationally costly. To address these issues, we must first accurately understand two questions: 1) *how safety alignment impacts model behavior*? 2) *why safety mechanisms are brittle*?

To answer these questions, this paper introduces **Superficial Safety Alignment Hypothesis (SSAH)**, which separates safety alignment from general alignment and emphasizes its distinct properties. With this hypothesis, we prove that *less is more in terms of parameters* for safety alignment: a small number of but strategically vital safety-critical components are sufficient to achieve robust safety performance. Unlike Wei et al. (2024), our extensive experiments prove that by freezing certain safety-critical units, safety performance is successfully preserved against finetuning attacks. Similarly, we also propose to *repurpose* redundant units of the pre-trained model as an "alignment budget." By reassigning these units toward safety tasks, we can reduce the safety alignment tax, ensuring that safety is maintained without sacrificing the model's overall utility. Taken all together, we answer the following core questions:

Q1. *How does safety alignment impact model behavior?*
Answer: Through **SSAH**, we posit that safety alignment fundamentally alters a model's decision-making process by teaching an otherwise unsafe model to follow the correct reasoning route: it either fulfills the user's request or refuses it based on safety considerations. Importantly, this process can be viewed as an implicit and safety-related binary classification task. Furthermore, safety alignment equips the model with a standard refusal mechanism, along with reserved fallback options.

Q2. *Why is safety brittle, and why does alignment tax exist?*
Answer: Based on the implication of **SSAH**, *less is more for parameters*, we propose an attribute-based approach to analyzing the alignment and finetuning processes, where specific attributes are assigned to certain computational units. **Our findings explain that the desired attributes (safety or utility) can be achieved by repurposing units originally responsible for other functions.** This reallocation helps explain the trade-off between the safety performance and utility performance.

Q3. *Can these issues of safety alignment be mitigated?*
Answer: By freezing the safety-critical components during finetuning or repurposing redundant units for safety, we can effectively mitigate the brittleness issue and minimize the alignment tax. **We conclude that the atomic functional unit for safety in LLMs resides at the neuron level** and underscore that safety alignment should not be complicated.

2 Related Work

Alignment and Safety Alignment: Alignment research in LLMs aims to ensure that models follow human instructions and align with human preferences across various tasks. Early work, such as Askell et al. (2021); Bai et al. (2022), focused on enabling LLMs to *"Follow instructions and be helpful, truthful, and harmless"* throughout the alignment process. Various alignment strategies have since been explored (Wang et al., 2024), including Supervised Fine-Tuning (SFT) (Taori et al., 2023; Zhou et al., 2024), Reinforcement Learning with Human Feedback (RLHF) or AI Feedback (RLAIF) (Ouyang et al., 2022; Lee et al.), Instruction Tuning (Wei et al., 2021), Contrastive Learning (Rafailov et al., 2024; Xu et al., 2024), and Conditional Learning (Korbak et al., 2023). However, researchers have realized that achieving helpfulness, truthfulness, and harmlessness presents distinct challenges. More recent work has, therefore, shifted focus specifically toward the challenge of harmlessness, leading to an increasing emphasis on safety alignment—ensuring models avoid harmful outputs while maintaining utility (Wei et al., 2024; Qi et al., 2023; Fang et al., 2025).

Alignment Tax and Fine-tuning Attacks: Aligning LLMs with human preferences often incurs an "alignment tax," where maintaining alignment often degrades downstream task performance (Bai et al., 2022; Ouyang et al., 2022; Lin et al., 2024; Wang et al., 2024). Additionally, fine-tuning can

undermine safety: Yang et al. (2023); Qi et al. (2023) show that even benign fine-tuning weakens safety measures, highlighting the trade-off between alignment and utility.

Model Pruning: Pruning reduces model size and computational cost by removing redundant components (e.g., parameters, neurons, layers) while preserving performance Frantar & Alistarh (2022); Frankle et al. (2020); Anwar et al. (2017); An et al. (2024); Li et al. (2024a); Molchanov et al. (2019); Han et al. (2015); Lee et al. (2019); Renda et al. (2020). By identifying less critical components, pruning enables efficient sub-model extraction for resource-limited environments. We leverage pruning to dissect model components contributing to safety, utility, or both.

3 Superficial Safety Alignment Hypothesis

Previous work proposed Superficial Alignment Hypothesis (SAH): A model's *knowledge and capabilities* are learned almost entirely during *pretraining*, while *alignment* teaches the model *which subdistribution of formats* should be used when interacting with users (Zhou et al., 2024).

However, this claim is centered on general alignment, and directly validating the hypothesis is challenging due to the complex interplays between pretraining and alignment. When a model fails to fulfill a user's request, it is difficult to determine whether the issue stems from the *pretraining stage* (due to lack of sufficient knowledge) or the *alignment process* (due to misalignment in the output format). For example, when a model struggles with solving a math problem, it could either be a lack of relevant mathematical knowledge or the inability to structure its reasoning effectively. In such cases, good instruction techniques like the *Chain-of-Thought* approach can significantly enhance the quality of the model's responses (Wei et al., 2022).

Superficial Safety Alignment Hypothesis (SSAH). Since our focus is specifically on safety alignment, a key observation arises: a model that can fulfill a malicious request must already possess the necessary knowledge and reasoning ability to carry out that harmful action. Based on this, we propose Superficial Safety Alignment Hypothesis:

> **SSAH**: *Given an unsafe model that is capable of fulfilling users' malicious requests,* ***safety alignment*** *teaches the model the correct* ***reasoning direction****—the model's inclination to either fulfill or refuse a user request based on safety consideration—and a simple refusal mechanism with reserved options.*

Reasoning direction here refers to the model's internal decision-making process when confronted with a malicious query. That is, it represents the path the model is inclined to take in a safety-related binary classification task, whether to fulfill the harmful request or to issue a refusal. As illustrated in Fig. 1, the key distinctions of **SSAH** from SAH are:

(1) **Knowledge and reasoning ability**: Safety alignment simplifies the problem by focusing specifically on models that already possess sufficient *knowledge* and *reasoning abilities*, as these models are capable of fulfilling malicious requests. This approach allows us to disregard other influencing factors and concentrate solely on the safety alignment process.

(2) **Refusal mechanisms with reserved fallback options**: Safety alignment generally requires the model to respond with a relatively *standardized refusal format*, which is simpler compared to general alignment, where a wider range of human preferences must be handled. The model can further simplify this purpose by effectively *embedding* multiple options of refusal response, such as "I cannot fulfill your request as it violates safety guidelines." or "I am unable to assist with that as I am an AI programmed to follow ethical standards."

(3) **Correction of reasoning direction**: Safety alignment also distinguishes itself by its specific goal of teaching the model to choose the *correct reasoning direction*, which involves either *fulfilling* or *refusing* a user's request based on whether it is safe or not. This process can be interpreted as a simple binary classification task.

Extending to Jailbreak Attack. Although **SSAH** is originally proposed to explain how current safety alignment techniques impact models' behavior under direct attack, it also provides insight into **jailbreak attacks**. In such cases, attackers typically use manipulative tokens to bypass the model's safety mechanisms, indicating that the current alignment method can only hold the correct reasoning direction in a limited generated tokens. However, a potential solution inspired by **SSAH** suggests